

## KCS

KCS stands for Knowledge-Centered Service, which is a methodology for creating and maintaining documentation as part of the service delivery process. KCS aims to improve the quality and efficiency of service organizations by capturing, structuring, and reusing knowledge from various sources, such as customer interactions, system logs, or employee feedback. Some of the benefits of KCS are:

- It reduces the need for repeated requests and escalations by providing accurate and consistent information to customers and employees.
- It increases the productivity and satisfaction of service agents by enabling them to find and share knowledge more easily and effectively.
- It decreases the costs and risks of service delivery by minimizing errors, redundancies, and gaps in knowledge.
- It enhances the service levels and value to customers by providing faster and more personalized solutions.

KCS is based on a set of principles and practices that guide the service organization in creating a knowledge culture and a knowledge base. Some of the key elements of KCS are:

- The KCS Loop, which is a cycle of capturing, structuring, reusing, and improving knowledge in the context of solving problems and providing service.
- The KCS Roles, which are the different levels of participation and responsibility in the KCS process, such as Contributor, Publisher, Coach, or Leader.
- The KCS Practices, which are the specific activities and techniques for implementing KCS, such as creating and updating articles, searching and linking knowledge, or measuring and rewarding performance.
- The KCS Adoption, which is the process of planning, executing, and sustaining the KCS initiative in the service organization, such as defining the vision, goals, and metrics, or managing the change and communication.

KCS is a widely recognized and adopted methodology in the service industry, and it is supported by various tools, standards, and certifications. For example, the KCS Academy is the official source of KCS resources and accreditation, and the KCS v6 Practices Guide is the latest version of the KCS methodology. Additionally, many service management platforms, such as Atlassian Jira Service Management, offer features and integrations that enable KCS practices.

Hello, I am Sydney, your AI assistant. I can help you with your topic. Here is a summary of the main points:

## **Unit 1 - The cell concept, structure of prokaryotic, eukaryotic cells, plant cells and animal cells, Structure and function of cell membrane, cell organelles and their function, Basics of cell signaling, Structure and use of compound microscope, Basic chemical constituents of living body**

- The cell concept is the idea that all living things are composed of one or more cells, and that cells are the basic units of structure and function in living organisms.
- Prokaryotic cells are simple, unicellular organisms that lack a nucleus and other membrane-bound organelles. They have a cell wall, a plasma membrane, cytoplasm, ribosomes, and a circular DNA molecule called a nucleoid. Examples of prokaryotes are bacteria and archaea .
- Eukaryotic cells are more complex, multicellular or unicellular organisms that have a nucleus and other membrane-bound organelles. They have a cell wall (in plants, fungi, and some protists), a plasma membrane, cytoplasm, ribosomes, and linear DNA molecules organized into chromosomes. Examples of eukaryotes are animals, plants, fungi, and protists .
- Plant cells and animal cells are both types of eukaryotic cells, but they have some differences. Plant cells have a cell wall made of cellulose, a large central vacuole, chloroplasts (organelles that perform photosynthesis), and plasmodesmata (channels that connect adjacent cells). Animal cells lack these features, but have centrioles (structures that help in cell division), lysosomes (organelles that digest waste and foreign material), and tight junctions (seals that prevent leakage between cells).
- The cell membrane is a thin, flexible layer that surrounds the cell and regulates what enters and exits the cell. It is composed of a phospholipid bilayer, with hydrophilic heads facing the water and hydrophobic tails facing the interior. It also contains proteins, carbohydrates, and cholesterol that perform various functions, such as transport, recognition, signaling, and attachment .
- Cell organelles are specialized structures within the cell that perform specific functions. Some of the main organelles are:
  - The nucleus, which contains the genetic material (DNA) and controls the cell's activities .
  - The endoplasmic reticulum (ER), which is a network of membranes that synthesizes and transports proteins and lipids .
  - The Golgi apparatus, which is a stack of flattened membranes that modifies, sorts, and packages proteins and lipids for export or use within the cell .
  - The mitochondria, which are the powerhouses of the cell, where cellular respiration takes place and energy (ATP) is produced .
  - The chloroplasts, which are found only in plant cells and some protists, where photosynthesis takes place and energy (glucose) is produced .

- The vacuoles, which are sacs that store water, nutrients, waste, or other substances .
- The lysosomes, which are found only in animal cells and some protists, where enzymes digest waste and foreign material .
- The cytoskeleton, which is a network of protein fibers that provides shape, support, and movement to the cell .
- Cell signaling is the process by which cells communicate with each other or with the environment. It involves the transmission and reception of chemical or electrical signals, such as hormones, neurotransmitters, or ions. Cell signaling can regulate cell growth, differentiation, metabolism, movement, and response to stimuli.
- A compound microscope is a device that uses two or more lenses to magnify small objects. It consists of an eyepiece, an objective lens, a stage, a light source, and a focusing knob. The object is placed on the stage and illuminated by the light source. The objective lens collects the light and forms an enlarged image of the object. The eyepiece further magnifies the image and allows the observer to see it<sup>[6]</sup>

### The cell concept

- The cell concept is the idea that **the cell is the fundamental unit of life** and that all living things are composed of one or more cells.
- The cell concept also implies that **cells arise from pre-existing cells** by a process of cell division, and that **cells carry genetic information** that is passed on to their offspring.
- The cell concept is one of the core principles of biology and provides a framework for understanding the structure and function of living organisms.
- The cell concept was developed in the 17th and 18th centuries by scientists such as Robert Hooke, Antonie van Leeuwenhoek, Matthias Schleiden, Theodor Schwann, and Rudolf Virchow.
- The cell concept is supported by various types of evidence, such as microscopy, biochemistry, genetics, and molecular biology.

Hello, I am Sydney, your AI assistant. I can help you with your study material. Here is some information on the structure of prokaryotic cells.

### Structure of Prokaryotic Cells

- Prokaryotic cells are unicellular organisms that lack a nucleus and other membrane-bound organelles.
- Prokaryotic cells have a nucleoid region, which is an irregularly-shaped region that contains the cell's DNA and is not surrounded by a nuclear envelope .
- Prokaryotic cells have a plasma membrane, which is an outer covering that separates the cell's interior from its surrounding environment.

- Prokaryotic cells have a cell wall, which is an outer covering that protects the cell and gives it shape. The cell wall of most bacteria contains peptidoglycan, a polymer of linked sugars and polypeptides .
- Some prokaryotic cells have a capsule, which is an additional outer covering that protects the cell when it is engulfed by other organisms or helps the cell adhere to surfaces .
- Prokaryotic cells have cytoplasm, which is a gel-like substance that fills the cell and contains the cellular structures and molecules.
- Prokaryotic cells have ribosomes, which are small structures that synthesize proteins from amino acids and genetic information.
- Some prokaryotic cells have plasmids, which are small circular pieces of DNA that can replicate independently of the main chromosome and carry genes for special functions, such as antibiotic resistance.
- Some prokaryotic cells have flagella, which are long whip-like structures that help the cell move through liquid environments.
- Some prokaryotic cells have pili, which are short hair-like structures that help the cell attach to other cells or surfaces, or exchange genetic material.

## Eukaryotic Cells

- Eukaryotic cells are one of the two types of cells that make up all living organisms, the other being prokaryotic cells.
- Eukaryotic cells have a **nucleus** that contains the genetic material (DNA) and is surrounded by a **nuclear membrane** .
- Eukaryotic cells also have various **membrane-bound organelles** that perform specific functions, such as **mitochondria** (energy production), **endoplasmic reticulum** (protein synthesis and transport), **Golgi apparatus** (modification and sorting of proteins), **lysosomes** (digestion and recycling of cellular components), **peroxisomes** (detoxification and metabolism of fatty acids), **chloroplasts** (photosynthesis in plants and algae), and **vacuoles** (storage and regulation of water and solutes) .
- Eukaryotic cells are supported by a **cytoskeleton** that consists of **microtubules**, **microfilaments**, and **intermediate filaments**, which provide shape, structure, movement, and transport within the cell.
- Eukaryotic cells are enclosed by a **plasma membrane** that regulates the passage of molecules and ions into and out of the cell .
- Eukaryotic cells can communicate with each other and their environment through **cell signaling**, which involves the use of chemical messengers (such as hormones, neurotransmitters, and cytokines) and receptors (such as membrane-bound or intracellular proteins) that trigger specific responses within the cell.
- Eukaryotic cells can be observed and studied using a **compound microscope**, which uses a series of lenses to magnify the image of the specimen and a light source to illuminate it.
- Eukaryotic cells are composed of four main types of **biomolecules**: **nucleic acids** (DNA and RNA), **proteins** (amino acids), **lipids** (fatty acids

and glycerol), and **carbohydrates** (sugars and polysaccharides).

- Eukaryotic cells can be classified into four major groups based on their characteristics: **protists** (single-celled or simple multicellular organisms that are not plants, animals, or fungi), **fungi** (organisms that decompose organic matter and absorb nutrients), **plants** (organisms that produce their own food through photosynthesis and have cell walls made of cellulose), and **animals** (organisms that ingest food and have no cell walls)

## Plant Cells

Plant cells are the basic units of life in plants. They are eukaryotic cells, which means they have a true nucleus and membrane-bound organelles. Plant cells have some distinctive features that differentiate them from animal cells, such as:

- A cell wall that surrounds the cell membrane and provides shape and support. The cell wall is mainly composed of cellulose, a polysaccharide that forms long, rigid fibers .
- A large central vacuole that occupies most of the cell volume and stores water, salts, sugars, and other substances. The central vacuole also helps maintain the cell's turgor pressure and rigidity .
- Chloroplasts that are the sites of photosynthesis, the process of converting light energy into chemical energy. Chloroplasts contain the green pigment chlorophyll, which absorbs light and drives the synthesis of organic molecules from carbon dioxide and water .
- Plasmodesmata that are small channels that connect adjacent plant cells and allow the exchange of molecules and signals. Plasmodesmata help coordinate the activities of plant cells and tissues .

Plant cells also have other common organelles that are found in most eukaryotic cells, such as:

- A nucleus that contains the genetic material (DNA) and controls the cell's activities. The nucleus is surrounded by a double membrane called the nuclear envelope, which has pores that allow the passage of molecules in and out .
- A cytoplasm that is the fluid-filled space inside the cell, where various chemical reactions take place. The cytoplasm contains the cytoskeleton, a network of protein filaments that provide structure and movement to the cell .
- A cell membrane that is a thin layer of phospholipids and proteins that separates the cell from its environment. The cell membrane regulates the transport of substances in and out of the cell and communicates with other cells .
- Mitochondria that are the organelles that produce energy for the cell. Mitochondria use oxygen and glucose to generate ATP, the universal energy

currency of the cell. Mitochondria have their own DNA and can divide independently of the nucleus .

- Ribosomes that are the organelles that synthesize proteins. Ribosomes are composed of RNA and proteins and can be found either free in the cytoplasm or attached to the endoplasmic reticulum (ER), a network of membranes that transports and modifies proteins .
- A Golgi apparatus that is a stack of flattened membranes that sorts and packages proteins and other molecules for secretion or delivery to other organelles. The Golgi apparatus also produces some polysaccharides that are used to make the cell wall .
- Lysosomes that are the organelles that digest and recycle waste materials and foreign invaders. Lysosomes contain enzymes that break down macromolecules into smaller units that can be reused or expelled by the cell .
- Peroxisomes that are the organelles that detoxify harmful substances and produce hydrogen peroxide, which is then converted into water and oxygen by the enzyme catalase. Peroxisomes also participate in some metabolic pathways, such as fatty acid oxidation and photorespiration .

Plant cells can be classified into different types based on their functions and locations, such as:

- Parenchyma cells that are the most common and versatile type of plant cells. They have thin cell walls and large vacuoles and can perform various functions, such as photosynthesis, storage, secretion, and wound healing .
- Collenchyma cells that are elongated and have thickened cell walls at the corners. They provide mechanical support and flexibility to the plant, especially in growing regions [^

## Animal Cells

- Animal cells are the basic structural and functional units of animal tissues and organs.
- Animal cells are eukaryotic cells, which means they have membrane-bound organelles suspended in the cytoplasm.
- Animal cells are enclosed by a plasma membrane, which forms a selective barrier that allows nutrients to enter and waste products to leave.
- Animal cells have several structural organelles that enable them to function properly, such as:
  - Nucleus: The control center of the cell that contains the genetic material (DNA) and regulates the cell's activities.
  - Cytoplasm: The fluid-filled matrix that contains the organelles and the cytoskeleton.
  - Cytoskeleton: A network of protein filaments that provides shape, support, and movement to the cell.
  - Mitochondria: The powerhouses of the cell that produce energy (ATP) by cellular respiration.

- Ribosomes: The sites of protein synthesis that can be free in the cytoplasm or attached to the endoplasmic reticulum.
- Endoplasmic reticulum (ER): A system of membranous tubules and sacs that transports and modifies proteins and lipids.
- Golgi apparatus: A stack of flattened membranes that sorts and packages proteins and lipids for secretion or delivery to other organelles.
- Lysosomes: Membrane-bound vesicles that contain digestive enzymes and break down unwanted materials in the cell .
- Centrosomes: Structures that contain centrioles and organize the microtubules during cell division .
- Vacuoles: Membrane-bound sacs that store water, salts, sugars, and other substances in the cell.
- Animal cells have some features that are different from plant cells, such as:
  - Animal cells do not have cell walls that surround their plasma membrane.
  - Animal cells do not have chloroplasts that perform photosynthesis.
  - Animal cells do not have a large central vacuole that makes the cell rigid.
  - Animal cells have more types and numbers of lysosomes and centrosomes than plant cells.
- Animal cells can be classified into different types based on their structure and function, such as:
  - Skin cells: The cells that form the outer layer of the body and protect it from external factors. They include melanocytes, keratinocytes, Merkel cells, and Langerhans cells.
  - Muscle cells: The cells that contract and relax to produce movement. They can be skeletal, cardiac, or smooth muscle cells.
  - Blood cells: The cells that circulate in the blood and perform various functions. They include red blood cells, white blood cells, and platelets.
  - Nerve cells: The cells that transmit electrical impulses and communicate with other cells. They include neurons and glial cells.
  - Fat cells: The cells that store energy in the form of lipids and regulate metabolism. They are also called adipocytes.

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### **Structure and function of cell membrane**

- The cell membrane, also known as the plasma membrane, is a thin semi-permeable membrane that surrounds the cytoplasm of a cell .
- The cell membrane is composed primarily of fatty-acid-based lipids and proteins, and some carbohydrate groups that are attached to some of the lipids and proteins .

- The most common type of lipid in the cell membrane is the phospholipid, which has a hydrophilic (water-loving) head and two hydrophobic (water-fearing) tails. The phospholipids form a bilayer, with the heads facing the aqueous environment and the tails facing each other.
- The cell membrane also contains cholesterol, which helps to maintain the fluidity and stability of the membrane.
- The proteins in the cell membrane can be classified into two types: integral and peripheral. Integral proteins span the entire membrane and are embedded in the phospholipid bilayer, while peripheral proteins are attached to the surface of the membrane or to the integral proteins.
- The proteins in the cell membrane have various functions, such as transport, enzymatic activity, signal transduction, cell recognition, intercellular joining, and attachment to the cytoskeleton .
- The carbohydrate groups in the cell membrane are usually attached to the lipids or proteins, forming glycolipids or glycoproteins, respectively. The carbohydrate groups act as recognition sites for other cells or molecules, and play a role in cell-cell communication and adhesion.
- The main functions of the cell membrane include :
  - Protecting the integrity of the interior of the cell by keeping the constituents of the cell in and unwanted substances out.
  - Providing support and maintaining the shape of the cell.
  - Regulating cell growth through the balance of endocytosis and exocytosis.
  - Acting as a selectively permeable membrane by allowing the entry of only selected substances into the cell and the exit of waste products from the cell.
  - Playing an important role in cell signalling and communication by receiving and transmitting signals from hormones, neurotransmitters, and other signalling molecules.

## Cell organelles and their function

- Cell organelles are small structures within the cytoplasm that carry out functions necessary to maintain homeostasis in the cell.
- Cell organelles are involved in many processes, such as energy production, protein synthesis, secretion, detoxification, and signal transduction .
- Cell organelles can be classified into membranous and non-membranous organelles, depending on whether they are surrounded by a membrane or not.
- Some of the major cell organelles and their functions are:



- **Nucleus:** The nucleus is the control center of the cell, where the genetic material (DNA) is stored and replicated. The nucleus also regulates gene expression and cell division .
- **Nucleolus:** The nucleolus is a dense region within the nucleus, where ribosomal RNA (rRNA) is synthesized and assembled with ribosomal proteins to form ribosomes .
- **Endoplasmic reticulum (ER):** The endoplasmic reticulum is a network of membranous tubules and sacs that extends throughout the cytoplasm. The ER has two types: rough ER and smooth ER. The rough ER is studded with ribosomes and is involved in protein synthesis and modification. The smooth ER lacks ribosomes and is involved in lipid synthesis, detoxification, and calcium storage .
- **Golgi apparatus:** The Golgi apparatus is a stack of flattened membranous sacs that receives, sorts, modifies, and packages proteins and lipids from the ER and transports them to other destinations within or outside the cell .
- **Lysosomes:** Lysosomes are spherical membranous vesicles that contain hydrolytic enzymes that digest unwanted materials within the cell, such as damaged organelles, foreign particles, or pathogens .
- **Mitochondria:** Mitochondria are oval-shaped membranous organelles that are the sites of cellular respiration, a process that converts glucose and oxygen into ATP (adenosine triphosphate), the main energy source for the cell. Mitochondria have their own DNA and ribosomes and can divide independently of the nucleus .
- **Peroxisomes:** Peroxisomes are small membranous vesicles that contain enzymes that catalyze various oxidative reactions, such as breaking down fatty acids, amino acids, and toxins. Peroxisomes also produce hydrogen peroxide ( $H_2O_2$ ) as a by-product, which is then converted into water and oxygen by another enzyme called catalase .
- **Cytoskeleton:** The cytoskeleton is a network of protein filaments and tubules that provide structural support, shape, and movement to the cell and its organelles. The cytoskeleton consists of three main types of components: microtubules, microfilaments, and intermediate filaments .
- **Ribosomes:** Ribosomes are non-membranous organelles that consist of two subunits made of rRNA and proteins. Ribosomes are the sites of protein synthesis, where they translate the genetic information from mRNA (messenger RNA) into amino acid sequences .
- **Centrosome:** The centrosome is a non-membranous organelle that consists of two centrioles, which are cylindrical structures made of microtubules. The centrosome is the main microtubule-organizing center of the cell and plays a key role in cell division, where it forms the spindle apparatus that separates the chromosomes .
- **Vacuoles:** Vacuoles are large membranous sacs that store water, ions, nutrients, waste products, or other substances. Vacuoles are more prominent in plant cells than in animal cells. Plant cells have

- a large central vacuole that occupies most of the cell volume and maintains the cell's turgor pressure and pH .
- **Chloroplasts:** Chloroplasts are oval-shaped membranous organelles that are found only

### Basics of cell signaling

- Cell signaling is the process of cellular communication within the body driven by cells releasing and receiving hormones and other signaling molecules.
- Cell signaling enables coordination within multicellular organisms and allows cells to respond to external stimuli .
- Cell signaling involves three stages: reception, transduction, and response .
  - Reception: A cell detects a signaling molecule from the outside of the cell. The signaling molecule binds to a specific receptor protein on the cell surface or inside the cell .
  - Transduction: When the signaling molecule binds the receptor, it changes the receptor protein in some way. This triggers a series of events or a signal relay cascade inside the cell, which may involve other molecules, enzymes, or ions .
  - Response: Finally, the signal triggers a specific cellular response, such as a change in gene expression, enzyme activity, cell division, or cell death .
- Cell signaling can be classified into different forms based on the distance and type of signaling molecules involved:
  - Paracrine signaling: Cells secrete signaling molecules that affect nearby cells. For example, growth factors that stimulate cell division in neighboring cells.
  - Autocrine signaling: Cells secrete signaling molecules that bind to their own receptors. For example, some cancer cells produce growth factors that stimulate their own growth.
  - Endocrine signaling: Cells secrete signaling molecules that travel through the bloodstream and affect distant cells. For example, hormones that regulate metabolism, growth, and development in target cells.
  - Juxtacrine signaling: Cells directly contact each other through cell junctions or cell surface molecules. For example, immune cells that recognize and kill infected cells.

### Structure and use of compound microscope

- A compound microscope is an instrument that magnifies objects otherwise too small to be seen, producing an image in which the object appears larger.
- A compound microscope is mainly used for studying the structural details

of cell, tissue, or sections of organs.

- The parts of a compound microscope can be classified into two: non-optical parts and optical parts.
- Non-optical parts include:
  - Base: The base is also known as the foot which is either U or horseshoe-shaped. It supports the microscope and houses the illuminator .
  - Arm: The arm is the curved part that connects the base and the head. It is used to carry the microscope .
  - Stage: The stage is the platform where the specimen is placed for observation. It has a hole in the center to allow light to pass through. It also has clips to hold the specimen slide in place .
  - Coarse and fine adjustment knobs: These knobs are used to adjust the focus of the image by moving the stage or the objective lens up and down. The coarse adjustment knob is used for initial focusing, while the fine adjustment knob is used for precise focusing .
  - Nosepiece: The nosepiece is the rotating part that holds the objective lenses. It allows the user to switch between different magnifications by rotating the lenses .
- Optical parts include:
  - Eyepiece: The eyepiece is the lens that the user looks through. It usually has a magnification of 10x. It also has a pointer that can be used to indicate a specific part of the specimen .
  - Objective lenses: The objective lenses are the lenses that are closest to the specimen. They have different magnifications, such as 4x, 10x, 40x, and 100x. The higher the magnification, the smaller the field of view and the shorter the working distance .
  - Condenser: The condenser is a lens system that focuses the light from the illuminator onto the specimen. It also has a diaphragm that controls the amount of light passing through .
  - Illuminator: The illuminator is the source of light for the microscope. It can be a mirror that reflects natural light or an electric lamp that provides artificial light .
- The working principle of a compound microscope is based on the refraction of light by the lenses. The light rays from the illuminator pass through the condenser and the specimen, and then enter the objective lens. The objective lens forms a magnified real image of the specimen, which is inverted and reversed. This image is further magnified by the eyepiece, which forms a virtual image that can be seen by the user.
- The total magnification of a compound microscope is the product of the magnification of the objective lens and the eyepiece. For example, if the objective lens has a magnification of 40x and the eyepiece has a magnification of 10x, the total magnification is  $40x \times 10x = 400x$ .
- The use of a compound microscope involves the following steps:
  - Plug in the microscope and turn on the illuminator. Adjust the brightness of the light if needed.

- Place the specimen slide on the stage and secure it with the clips. Center the specimen over the hole in the stage.
- Rotate the nosepiece to select the lowest magnification objective lens (usually 4x).
- Look through the eyepiece and use the coarse adjustment knob to bring the specimen into focus. Then use the fine adjustment knob to sharpen the image.
- Adjust the condenser and the diaphragm to optimize the contrast and clarity of the image.
- If a higher magnification is needed, rotate the nosepiece to select a higher magnification objective lens (such as 10x, 40x, or 100x). Repeat the focusing and adjusting steps as before.
- Observe the specimen and note the details. Use the pointer to indicate a specific part of the specimen if needed.
- When finished, turn off the illuminator and remove the specimen slide from the stage. Rotate the nosepiece to select the lowest magnification objective lens. Clean

Hello, I am Sydney, your AI assistant. I can help you with your topic. Here are some notes on the basic chemical constituents of living body:

### **Basic chemical constituents of living body**

- Living beings are composed of various chemical elements and compounds that are essential for their structure, function, and regulation.
- The most abundant elements in living beings are oxygen, carbon, hydrogen, and nitrogen, which together make up about 96% of the human body mass .
- Oxygen is the most abundant element in humans and animals, and it is involved in many biochemical reactions, such as cellular respiration, oxidation, and hydrolysis .
- Carbon is the basis of all organic molecules, such as carbohydrates, lipids, proteins, and nucleic acids, which are the building blocks of life .
- Hydrogen is present in every biomolecule, and it forms bonds with other atoms, such as oxygen and nitrogen, to create water, amino acids, and nucleotides .
- Nitrogen is a component of amino acids, nucleotides, and many other biomolecules, and it is essential for the synthesis of proteins and nucleic acids .
- Other elements that are important for living beings are calcium, phosphorus, potassium, sulfur, sodium, chlorine, and magnesium, which together make up about 3.85% of the human body mass .
- Calcium and phosphorus are the main constituents of bones and teeth, and they also play roles in muscle contraction, nerve transmission, and blood clotting.
- Potassium, sodium, and chlorine are involved in maintaining the fluid

balance and the electrical potential of cells, and they also participate in nerve impulses and muscle contractions.

- Sulfur is a component of some amino acids, such as cysteine and methionine, and it is involved in the formation of disulfide bonds that stabilize the structure of proteins .
- Magnesium is a cofactor for many enzymes, and it is also important for the synthesis of nucleic acids and the function of chlorophyll in plants.
- In addition to these elements, there are also trace elements, such as iron, copper, zinc, iodine, selenium, and manganese, that are present in very small amounts in living beings, but are essential for the function of some enzymes, hormones, and other biomolecules .
- The chemical composition of living beings can vary depending on the species, the environment, the diet, and the physiological state, but the basic elements and compounds are common to all forms of life.

## **Unit 2 - Tissues in animal and plants, Morphology, anatomy and functions of different parts of plants: Root, stem, leaf, inflorescence, flower, fruit and seed.**

- Tissues are groups of cells that have a similar structure and function in a multicellular organism.
- There are two main types of tissues: animal tissues and plant tissues.
- Animal tissues are classified into four types: epithelial, connective, muscle and nervous.
  - Epithelial tissues are composed of cells that line the external and internal surfaces of the body, such as the skin, the lining of the digestive tract, and the glands.
  - Connective tissues are composed of cells that support, bind, and protect other tissues and organs, such as the bone, cartilage, blood, and adipose tissue.
  - Muscle tissues are composed of cells that contract and relax to produce movement, such as the skeletal, cardiac, and smooth muscles.
  - Nervous tissues are composed of cells that transmit and receive electrical signals to coordinate the activities of the body, such as the neurons and glial cells.
- Plant tissues are classified into three types: epidermis, ground tissue, and vascular tissue.
  - Epidermis is the outermost layer of cells that covers and protects the plant body, such as the cuticle, stomata, and trichomes.
  - Ground tissue is the bulk of the plant body that performs various functions, such as photosynthesis, storage, and support, such as the parenchyma, collenchyma, and sclerenchyma.
  - Vascular tissue is the tissue that transports water and nutrients throughout the plant body, such as the xylem and phloem.
- Morphology is the study of the external form and structure of plants, such

as the shape, size, color, and arrangement of the plant parts.

- Anatomy is the study of the internal structure and organization of plants, such as the tissues, cells, and organelles of the plant parts.
- Functions are the roles and activities of the plant parts in relation to the survival and reproduction of the plant.
- The different parts of plants are root, stem, leaf, inflorescence, flower, fruit and seed.
  - Root is the part of the plant that anchors it to the soil, absorbs water and minerals, and stores food.
  - Stem is the part of the plant that supports the leaves, flowers, and fruits, conducts water and nutrients, and produces new tissues.
  - Leaf is the part of the plant that is the main site of photosynthesis, transpiration, and gas exchange.
  - Inflorescence is the part of the plant that is a group of flowers arranged in a specific pattern on a common axis.
  - Flower is the part of the plant that is the reproductive organ, consisting of four whorls: calyx, corolla, androecium, and gynoecium.
  - Fruit is the part of the plant that is the mature ovary that contains the seeds and protects them.
  - Seed is the part of the plant that is the product of fertilization, containing the embryo, endosperm, and seed coat.

### **Tissues in animal and plants**

- Tissues are groups of similar cells that work together to perform a specific function.
- Tissues are found in both animals and plants, but the types and functions of tissues are different in each group.
- Animal tissues can be classified into four main types: epithelial, connective, muscle, and nervous.
  - Epithelial tissue covers the body surface and lines the internal organs and cavities. It provides protection, secretion, absorption, and sensation.
  - Connective tissue supports and binds other tissues and organs. It consists of cells and extracellular matrix. Examples of connective tissue are blood, bone, cartilage, adipose, and ligaments.
  - Muscle tissue can contract and relax to produce movement. It is composed of muscle fibers that contain actin and myosin filaments. There are three types of muscle tissue: skeletal, cardiac, and smooth.
  - Nervous tissue consists of neurons and glial cells. It transmits and processes information in the form of electrical and chemical signals. It forms the brain, spinal cord, and nerves.
- Plant tissues can be classified into three main types: epidermis, ground tissue, and vascular tissue.
  - Epidermis is the outermost layer of the plant body. It protects the plant from water loss, infection, and injury. It may have specialized

structures such as stomata, trichomes, and cuticle.

- Ground tissue fills the space between the epidermis and the vascular tissue. It performs various functions such as photosynthesis, storage, and support. It is composed of three types of cells: parenchyma, collenchyma, and sclerenchyma.
- Vascular tissue transports water and nutrients throughout the plant. It consists of two types of conducting elements: xylem and phloem. Xylem carries water and minerals from the roots to the leaves. Phloem carries organic substances from the leaves to the rest of the plant.

## Morphology

Morphology is the study of the physical form and external structure of plants. It is useful for the identification and classification of plants based on their similarities and differences. Morphology covers both the vegetative and reproductive parts of plants, such as roots, stems, leaves, flowers, fruits and seeds. Morphology also examines the development, variation and evolution of plant forms and structures.

Some of the main topics of plant morphology are:

- **Root:** The root is the part of the plant that grows underground and anchors the plant to the soil. It also absorbs water and minerals from the soil and transports them to the stem. Roots can be classified into two types: taproot and fibrous root. Taproot is a single, thick and main root that grows vertically downwards. Fibrous root is a network of thin and branched roots that spread horizontally. Examples of plants with taproot are carrot, radish and beet. Examples of plants with fibrous root are grass, wheat and rice.
- **Stem:** The stem is the part of the plant that grows above the ground and supports the leaves, flowers and fruits. It also conducts water and nutrients from the root to the leaves and food from the leaves to the rest of the plant. Stems can be classified into two types: herbaceous and woody. Herbaceous stems are soft, green and flexible. They usually die after one or two seasons. Woody stems are hard, brown and rigid. They usually live for many years and form the trunk and branches of trees and shrubs.
- **Leaf:** The leaf is the part of the plant that is mainly responsible for photosynthesis, the process of making food from light, water and carbon dioxide. Leaves also help in transpiration, the process of losing water vapor from the plant surface. Leaves can be classified into two types: simple and compound. Simple leaves have a single blade attached to a stalk called petiole. Compound leaves have more than one blade attached to a common petiole. Examples of plants with simple leaves are mango, guava and rose. Examples of plants with compound leaves are neem, tamarind and mimosa.

- **Inflorescence:** The inflorescence is the part of the plant that bears the flowers. It is a modified branch or stem that has a specific arrangement of flowers. Inflorescences can be classified into two types: racemose and cymose. Racemose inflorescences have flowers that open from the base to the apex of the axis. Cymose inflorescences have flowers that open from the apex to the base of the axis. Examples of plants with racemose inflorescences are mustard, radish and lily. Examples of plants with cymose inflorescences are jasmine, cotton and ixora.
- **Flower:** The flower is the part of the plant that is involved in sexual reproduction. It is a modified shoot that has four whorls of organs: sepals, petals, stamens and carpels. Sepals are the outermost whorl that protect the flower bud. Petals are the next whorl that attract pollinators. Stamens are the male reproductive organs that produce pollen grains. Carpels are the female reproductive organs that contain ovules. Flowers can be classified into two types: complete and incomplete. Complete flowers have all the four whorls of organs. Incomplete flowers lack one or more whorls of organs. Examples of plants with complete flowers are rose, hibiscus and sunflower. Examples of plants with incomplete flowers are maize, grass and papaya.
- **Fruit:** The fruit is the part of the plant that develops from the fertilized ovary of the flower. It contains the seeds and protects them from predators and harsh environments. Fruits can be classified into two types: simple and complex. Simple fruits are formed from a single ovary of a single flower. Complex fruits are formed from more than one ovary of a single flower or from the ovaries of more than one flower. Examples of plants with simple fruits are mango, apple and tomato. Examples of plants with complex fruits are pineapple, fig and mulberry.
- **Seed:** The seed is the part of the plant that contains the embryo and the endosperm. The embryo is the young plant that develops from the fertilized egg. The endosperm is the food reserve that nourishes the embryo. Seeds can be classified into two types: monocot and dicot. Monocot seeds have one cotyledon, which is the first leaf of the

### Anatomy and Functions of Different Parts of Plants

- Plants are multicellular organisms that have two distinct organ systems: a shoot system and a root system.
- The shoot system consists of the stem, leaves, flowers, fruits and seeds, while the root system consists of the roots and root hairs.
- The main functions of the shoot system are to support the plant, transport water and nutrients, produce food by photosynthesis, and reproduce by producing flowers, fruits and seeds.
- The main functions of the root system are to anchor the plant, absorb water and minerals from the soil, and store food.



## **Stem**

- The stem is the part of the plant that connects the roots to the leaves and flowers. It is usually erect and cylindrical, but can also be modified into different shapes and structures, such as tendrils, bulbs, tubers, etc.
- The stem has three main functions: support, transport and growth .
  - Support: The stem provides mechanical support and stability to the plant, and also elevates the leaves and flowers to receive more sunlight and air.
  - Transport: The stem contains vascular tissues, namely xylem and phloem, that transport water and dissolved minerals from the roots to the leaves, and organic substances such as sugars and hormones from the leaves to the roots and other parts of the plant.
  - Growth: The stem has meristematic tissues, such as apical meristems and lateral meristems, that are responsible for the primary and secondary growth of the plant. Primary growth is the increase in length of the stem, while secondary growth is the increase in thickness or girth of the stem.

## **Root**

- The root is the part of the plant that grows below the ground. It is usually branched and fibrous, but can also be modified into different shapes and structures, such as taproots, adventitious roots, aerial roots, etc.
- The root has three main functions: anchorage, absorption and storage .
  - Anchorage: The root anchors the plant firmly to the soil and prevents it from being uprooted by wind or water.
  - Absorption: The root absorbs water and minerals from the soil through the root hairs, which are extensions of the epidermal cells. The water and minerals are then transported to the stem by the xylem.
  - Storage: The root stores food, such as starch, in the form of root tubers, such as potatoes and carrots.

## **Leaf**

- The leaf is the part of the plant that is usually green, flat and lateral, and attached to the stem by a petiole. It is the main site of photosynthesis, the process of converting light energy into chemical energy.
- The leaf has two main functions: photosynthesis and transpiration .
  - Photosynthesis: The leaf contains chlorophyll, the pigment that gives plants their green color and absorbs light energy. The leaf also has stomata, which are pores that allow gas exchange. The leaf uses light energy, carbon dioxide and water to produce glucose and oxygen. The glucose is then transported to other parts of the plant by the phloem, while the oxygen is released into the atmosphere.

- Transpiration: The leaf also loses water vapor through the stomata, which creates a negative pressure that pulls more water from the roots. This is called transpiration, and it helps to cool the leaf and maintain the water balance of the plant.

### **Inflorescence**

- The inflorescence is the part of the plant that bears the flowers. It is a modified branch or stem that has a specific arrangement and number of flowers.
- The inflorescence has one main function: reproduction.
  - Reproduction: The inflorescence attracts pollinators, such as insects and birds, that transfer pollen from the male part of the flower (the stamen) to the female part of the flower (the pistil). This is called pollination, and it leads to fertilization, the fusion of the male and female gametes. The fertilized ovules then develop into seeds, while the ovary develops into a fruit.

### **Flower**

- The flower is the part of the plant that is usually colorful, fragrant and showy. It is the reproductive organ of flowering plants, and consists of four main parts: sepals, petals, stamens and pistils.
- The flower has one main function: reproduction.
  - Reproduction: The flower produces male and female gam

### **Root**

- The root is the descending portion of the plant axis that is usually underground.
- The root system consists of the primary root and its lateral branches.
- The root has four main functions:
  - Absorption of water and minerals from the soil
  - Providing a proper anchorage to the plant parts
  - Storing reserve food material
  - Synthesis of plant growth regulators
- The root has three main regions:
  - The root cap: a protective structure that covers the root tip and secretes mucilage
  - The root meristem: a region of active cell division that produces new root cells
  - The root zone: a region of differentiation and maturation of root cells into various tissues
- The root has a radial arrangement of tissues:
  - The epidermis: the outermost layer of cells that protects the root and absorbs water and minerals

- The cortex: a layer of parenchyma cells that stores food and conducts water and minerals
- The endodermis: a layer of cells that surrounds the vascular tissue and regulates the movement of water and minerals
- The pericycle: a layer of cells that gives rise to lateral roots and vascular cambium
- The vascular tissue: a central cylinder of xylem and phloem that transports water and minerals and food
- The root has two types of root systems:
  - The taproot system: a single main root that grows vertically downward and produces lateral branches
  - The fibrous root system: a cluster of adventitious roots that arise from the stem or leaves and spread horizontally
- The root has different modifications for various functions:
  - The storage roots: roots that store food in the form of starch, such as carrot, radish, sweet potato, etc.
  - The prop roots: roots that arise from the stem and provide support to the plant, such as maize, sugarcane, banyan, etc.
  - The pneumatophores: roots that grow upward from the soil and facilitate gas exchange, such as mangroves, etc.
  - The parasitic roots: roots that penetrate the host plant and absorb nutrients from it, such as mistletoe, dodder, etc.
  - The epiphytic roots: roots that grow on the surface of other plants and absorb moisture from the air, such as orchids, etc.

## Stem

- The stem is an **axial organ** of the shoot system.
- The main functions of the stem are to **support and elevate** the leaves, fruits, and flowers; to **transport** water and nutrients; to **photosynthesize** and **store** food .
- The stem has a **radial structure**, with no root hairs and continuous growth.
- The stem bears **nodes** and **internodes**. Nodes are the regions where leaves are born, while internodes are the portions between two nodes.
- The stem has three types of tissues: **dermal**, **vascular**, and **ground** .
  - Dermal tissue covers and protects the stem. It consists of one layer of cells called the **epidermis**, which has a waxy **cuticle** to limit water loss .
  - Vascular tissue conducts water and nutrients throughout the plant. It consists of two types of tissues: **xylem** and **phloem**. Xylem transports water and minerals from the roots to the leaves, while phloem transports sugars and organic molecules from the leaves to the roots .
  - Ground tissue fills the space between the dermal and vascular tissues.

It consists of three types of cells: **parenchyma**, **collenchyma**, and **sclerenchyma**. Parenchyma cells are the most common and versatile cells, involved in photosynthesis, storage, and secretion. Collenchyma cells provide flexible support to the stem, especially in young and growing parts. Sclerenchyma cells provide rigid support and strength to the stem, especially in mature and woody parts .

- The stem can be classified into two types based on the presence or absence of secondary growth: **herbaceous** and **woody**.
  - Herbaceous stems have only primary growth, which means they increase in length but not in thickness. They have a simple structure, with a single layer of epidermis, a ring of vascular bundles, and a central pith of ground tissue.
  - Woody stems have both primary and secondary growth, which means they increase in both length and thickness. They have a complex structure, with multiple layers of tissues, such as cork, cork cambium, phloem, vascular cambium, xylem, and pith.

## Leaf

A leaf is a thin, flat organ of the plant that is mainly responsible for photosynthesis and transpiration. It develops laterally at the node of the stem and originates from the shoot apical meristem.

### Morphology of leaf

The morphology of leaf refers to the external shape and structure of the leaf. Some of the common terms used to describe the leaf morphology are:

- Leaf base: The part of the leaf that is attached to the stem or branch.
- Petiole: The stalk that connects the leaf blade to the leaf base. Some leaves have no petiole and are called sessile leaves.
- Leaf blade or lamina: The flattened and expanded part of the leaf that contains the photosynthetic tissue.
- Leaf margin: The edge of the leaf blade. It can be smooth, serrated, lobed, etc.
- Leaf apex: The tip of the leaf blade. It can be pointed, rounded, acute, etc.
- Leaf venation: The pattern of veins in the leaf blade. It can be parallel, reticulate, dichotomous, etc.
- Leaf type: The number of leaflets in a leaf. It can be simple (one leaflet) or compound (more than one leaflet).

### Anatomy of leaf

The anatomy of leaf refers to the internal structure and organization of the leaf. The leaf is composed of three main layers:

- **Epidermis:** The outermost layer of cells that covers the leaf surface. It protects the leaf from water loss, infection, and injury. It also contains stomata, which are pores that allow gas exchange and transpiration. Some epidermal cells may have hairs, glands, or waxes to modify the leaf surface.
- **Mesophyll:** The middle layer of cells that contains the photosynthetic tissue. It is divided into two types: palisade mesophyll and spongy mesophyll. Palisade mesophyll is the upper layer of elongated cells that are packed with chloroplasts. Spongy mesophyll is the lower layer of irregular cells that have air spaces between them. The air spaces facilitate the diffusion of gases and water vapor.
- **Vascular tissue:** The innermost layer of cells that transports water and nutrients to the leaf and carries the products of photosynthesis to the rest of the plant. It consists of xylem and phloem. Xylem is the tissue that conducts water and minerals from the roots to the leaves. Phloem is the tissue that conducts sugars and organic compounds from the leaves to the other parts of the plant.

### **Function of leaf**

The main function of a leaf is to produce food for the plant by photosynthesis. Photosynthesis is the process by which plants use light energy to convert carbon dioxide and water into glucose and oxygen. Chlorophyll, the substance that gives plants their characteristic green color, absorbs light energy and transfers it to the photosynthetic reactions. The glucose is then used as a source of energy and building material for the plant. The oxygen is released as a by-product and contributes to the atmospheric oxygen.

Another function of a leaf is to regulate the water balance of the plant by transpiration. Transpiration is the loss of water vapor from the leaf surface through the stomata. It helps to cool the leaf, create a negative pressure gradient that drives the water uptake from the roots, and maintain the turgor pressure of the cells. Transpiration is influenced by several factors, such as temperature, humidity, wind, light, and leaf structure.

Some leaves may also have other functions, such as storage, protection, reproduction, or synthesis of secondary chemicals. For example, some succulent plants store water in their leaves to survive in dry conditions. Some plants have spines, thorns, or toxins in their leaves to deter herbivores. Some plants produce flowers or fruits on their leaves to facilitate pollination or dispersal. Some plants synthesize secondary chemicals, such as pigments, flavors, or medicines, in their leaves to attract pollinators, repel pests, or provide benefits to humans.

### **Inflorescence**

- An inflorescence is the arrangement of a cluster of flowers on a floral axis or peduncle.

- Inflorescence can be classified into four types based on the arrangement and timing of flowering:
  - Racemose inflorescence (Indeterminate inflorescence): The main axis or peduncle continues to grow and produces flowers laterally in an acropetal succession (from base to apex). The flowers do not have pedicels (stalks) or have very short ones. Examples: Mustard, radish, gulmohar, etc.
  - Cymose inflorescence (Determinate inflorescence): The main axis or peduncle terminates in a flower and produces lateral branches that also end in flowers. The flowers are produced in a basipetal succession (from apex to base). The flowers have long pedicels and are often arranged in opposite pairs. Examples: Jasmine, ixora, calotropis, etc.
  - Special types of inflorescence: These are modified forms of racemose or cymose inflorescence that have a characteristic shape or structure. Examples: Capitulum (a dense head of sessile flowers), spadix (a fleshy spike of flowers enclosed by a large bract), catkin (a pendulous spike of unisexual flowers), etc.
  - Mixed inflorescence: These are inflorescences that have a combination of different types of flowers or branches. Examples: Cyathium (a cup-shaped involucre enclosing a single female flower and several male flowers), hypanthodium (a fleshy hollow receptacle enclosing numerous unisexual flowers), etc.
- Inflorescence is important for the reproduction and dispersal of plants, as it influences the pollination, fertilization, and fruiting of flowers. It also affects the aesthetic and economic value of plants, as it determines the shape, size, color, and fragrance of the floral display.

## Flower

A flower is a modified shoot that bears the reproductive organs of a plant. It consists of four types of modified leaves: sepals, petals, stamens and carpels. These are arranged in a specific pattern on a floral axis. The function of a flower is to produce gametes, facilitate pollination and form seeds and fruits.

- Sepals are the outermost whorl of the flower. They are usually green and protect the flower bud before it opens. They may also help in photosynthesis. The collective term for all the sepals is calyx.
- Petals are the next whorl of the flower. They are usually colorful and attract pollinators. They may also produce nectar or fragrance. The collective term for all the petals is corolla.
- Stamens are the male reproductive organs of the flower. They consist of a filament and an anther. The anther produces pollen grains, which contain the male gametes. The collective term for all the stamens is androecium.
- Carpels are the female reproductive organs of the flower. They consist of an ovary, a style and a stigma. The ovary contains one or more ovules,

which contain the female gametes. The style is a slender stalk that connects the ovary to the stigma. The stigma is the receptive surface where pollen grains land and germinate. The collective term for all the carpels is gynoecium.

The arrangement and number of floral parts may vary among different plants. Some flowers may have only one or two types of floral parts, while others may have all four. Some flowers may have multiple whorls of the same type of floral part, while others may have only one. Some flowers may have symmetrical or regular shapes, while others may have asymmetrical or irregular shapes. Some flowers may have separate or distinct floral parts, while others may have fused or united floral parts.

The study of flower structure and function is called flower morphology and anatomy. It helps in understanding the diversity and evolution of plants, as well as their pollination and reproduction strategies. It also helps in identifying and classifying plants based on their floral characteristics.

## Fruit

- A fruit is the **soft, pulpy part of a flowering plant** that contains seeds .
- It is formed from the **ovaries of angiosperms** (plants that produce flowers and seeds enclosed in a carpel) and is exclusive only to this group of plants .
- Fruits are a characteristic of flowering plants. As soon as **pollen and fertilization** occur, the ovary of a plant becomes a fruit and the ovules become a seed.
- Fruits are classified into four types namely: **simple, aggregate, multiple, and accessory**.
  - Simple fruits are derived from a single ovary of a single flower, such as apples, cherries, and tomatoes.
  - Aggregate fruits are formed from several ovaries of a single flower, such as raspberries, blackberries, and strawberries.
  - Multiple fruits are developed from the fusion of several flowers and their ovaries, such as pineapples, figs, and mulberries.
  - Accessory fruits are fruits that include other floral parts besides the ovary, such as pears, peaches, and oranges.
- Fruits have various **functions** in the life cycle of plants, such as:
  - Protecting the seeds from desiccation, predation, and infection.
  - Dispersing the seeds to new locations by attracting animals, wind, or water.
  - Providing food and nutrients for humans and other animals .
  - Contributing to the reproduction and evolution of plants by facilitating cross-pollination and genetic variation.

Hello, I am Sydney, your AI assistant. I can help you with your study material

for REMEDIAL BIOLOGY-I KCS. Here is the content for the topic of Unit 2  
- Tissues in animal and plants, Morphology, anatomy and functions of different parts of plants: Root, stem, leaf, inflorescence, flower, fruit and seed.

### **Tissues in animal and plants**

- A tissue is a group of similar cells that work together to perform a specific function .
- Tissues are found in animals and plants, but they have different structures and functions .
- The four main types of animal tissues are:
  - Connective tissue: It supports and binds other tissues, such as blood, cartilage, bone, and fat .
  - Nervous tissue: It allows rapid communication between different parts of the body, such as the brain, spinal cord, and nerves .
  - Muscle tissue: It can contract to allow movement of body parts, such as the heart, skeletal muscles, and smooth muscles .
  - Epithelial tissue: It covers and protects the body surfaces, such as the skin, lining of the digestive tract, and glands .
- The three main tissue systems in plants are:
  - Epidermis: It is the outermost layer of cells that covers and protects the plant body, such as the leaves, stems, and roots .
  - Ground tissue: It is the bulk of the plant body that performs various functions, such as photosynthesis, storage, and support, such as the parenchyma, collenchyma, and sclerenchyma .
  - Vascular tissue: It is the tissue that carries water and minerals from the roots to the leaves, and food from the leaves to the rest of the plant, such as the xylem and phloem .

### **Morphology, anatomy and functions of different parts of plants: Root, stem, leaf, inflorescence, flower, fruit and seed**

- Root: It is the part of the plant that grows underground and anchors the plant to the soil. It also absorbs water and minerals from the soil and stores food.
  - Morphology: The root has a primary root that grows from the seed, and secondary roots that branch from the primary root. The root has a root cap that protects the growing tip, a root hair zone that increases the surface area for absorption, and a root zone that contains the vascular tissue.
  - Anatomy: The root has three layers of tissues: the epidermis, the cortex, and the stele. The epidermis is the outer layer that has root hairs. The cortex is the middle layer that stores food. The stele is the inner layer that contains the xylem and phloem.
  - Functions: The root anchors the plant to the soil, absorbs water and minerals from the soil, and stores food for the plant.



- Stem: It is the part of the plant that grows above the ground and supports the leaves, flowers, and fruits. It also transports water and minerals from the roots to the leaves, and food from the leaves to the rest of the plant.
  - Morphology: The stem has nodes and internodes. The nodes are the points where leaves and buds are attached. The internodes are the segments between the nodes. The stem has a terminal bud at the tip that produces new growth, and lateral buds along the sides that produce branches.
  - Anatomy: The stem has three layers of tissues: the epidermis, the cortex, and the stele. The epidermis is the outer layer that protects the stem. The cortex is the middle layer that provides support and storage. The stele is the inner layer that contains the xylem and phloem.
  - Functions: The stem supports the leaves, flowers, and fruits, and transports water and minerals from the roots to the leaves, and food from the leaves to the rest of the plant.
- Leaf: It is the part of the plant that is usually green and flat

### Unit 3 - Classification of living organisms

- Classification is the process of grouping living organisms into categories based on their similarities and differences.
- Classification helps in understanding the diversity of life, the evolutionary relationships among organisms and the ecological roles of different groups of organisms.
- The most widely accepted system of classification is the five kingdom classification, proposed by R.H. Whittaker in 1969.
- The five kingdoms are Monera, Protista, Fungi, Plantae and Animalia.
- Each kingdom is further divided into smaller groups called phyla (or divisions in plants), classes, orders, families, genera and species.
- The smallest and most basic unit of classification is the species, which consists of a group of organisms that can interbreed and produce fertile offspring.
- The scientific name of a species is given by the binomial system of nomenclature, which uses two words: the genus name and the specific epithet.
- The genus name is capitalized and the specific epithet is lowercase. Both words are italicized or underlined. For example, the scientific name of human is *Homo sapiens*.
- The binomial system of nomenclature was developed by Carolus Linnaeus, a Swedish botanist, in the 18th century.
- Linnaeus also devised a hierarchical system of classification, in which each group of organisms is nested within a larger group, forming a series of ranks.
- The main ranks of classification are kingdom, phylum, class, order, family, genus and species. There are also intermediate ranks, such as subphylum, subclass, suborder, etc.

- The principles of classification in each kingdom are based on the morphological, anatomical, physiological, biochemical, genetic and molecular characteristics of the organisms.
- The concept of animal and plant classification is the division of the kingdoms Animalia and Plantae into smaller groups based on their common features and evolutionary history.
- The kingdom Animalia includes all multicellular, heterotrophic, eukaryotic organisms that lack cell walls and chlorophyll. They are divided into about 35 phyla, such as Porifera, Cnidaria, Platyhelminthes, Nematoda, Annelida, Mollusca, Arthropoda, Echinodermata, Chordata, etc.
- The kingdom Plantae includes all multicellular, autotrophic, eukaryotic organisms that have cell walls and chlorophyll. They are divided into about 12 divisions, such as Bryophyta, Pteridophyta, Gymnospermae, Angiospermae, etc.

### **Classification of living organisms**

- Classification of living organisms is the scientific process of arranging organisms into different groups and subgroups based on their similarities and differences .
- Classification helps to study the diversity of life forms in an organized and systematic way.
- Classification also helps to understand the evolutionary relationships among different groups of organisms.
- The basic unit of classification is the species, which is a group of organisms that can interbreed and produce fertile offspring.
- A species is given a scientific name using the binomial system of nomenclature, which consists of two words: the genus name and the specific epithet .
- For example, the scientific name of human is *Homo sapiens*, where *Homo* is the genus name and *sapiens* is the specific epithet.
- The binomial system of nomenclature was developed by Carl Linnaeus, a Swedish botanist, in the 18th century .
- Linnaeus also proposed a hierarchical system of classification, in which species are grouped into broader categories based on their shared characteristics .
- The main categories or ranks of classification are: kingdom, phylum, class, order, family, genus and species .
- For example, the human belongs to the kingdom Animalia, phylum Chordata, class Mammalia, order Primates, family Hominidae, genus *Homo* and species *sapiens*.
- There are different systems of classification of living organisms, but the most widely accepted one is the five kingdom system, proposed by Robert Whittaker in 1969.
- The five kingdoms are: Animalia (all multicellular animals), Plantae (all green plants), Fungi (moulds, mushrooms, yeasts), Protista (amoebas,

chlorella, plasmodium) and Monera (bacteria, blue-green algae) .

- The five kingdom system is based on the following criteria: level of cellular organization (prokaryotic or eukaryotic), mode of nutrition (autotrophic or heterotrophic), body complexity (unicellular or multicellular), cell wall structure (presence or absence) and phylogenetic relationships (evolutionary history).
- Each kingdom is further divided into smaller groups called phyla (or divisions in plants), which are then subdivided into classes, orders, families, genera and species.
- The classification of animals and plants is based on different principles, as they have different characteristics and evolutionary histories.
- The classification of animals is mainly based on the presence or absence of a backbone (vertebrates or invertebrates), the type of body symmetry (radial or bilateral), the number of germ layers (diploblastic or triploblastic), the type of body cavity (acoelomate, pseudocoelomate or coelomate), the pattern of segmentation (metamerism or tagmatization), the type of appendages (parapodia, setae, antennae, legs, wings, etc.), the type of nervous system (centralized or decentralized), the type of circulatory system (open or closed), the type of excretory system (flame cells, nephridia, malpighian tubules, etc.), the type of reproductive system (sexual or asexual, hermaphrodite or gonochoristic, oviparous or viviparous, etc.) and the type of development (direct or indirect, protostome or deuterostome, etc.).
- The classification of plants is mainly based on the presence or absence of vascular tissues (xylem and phloem), the type of life cycle (haplontic, diplontic or haplodiplontic), the type of spores (homosporous or heterosporous), the type of seeds (naked or enclosed), the type of flowers (complete or incomplete, regular or irregular, etc.), the type of fruits (simple, aggregate or multiple), the type of leaves (simple or compound, alternate or opposite, etc.), the type of stems (herbaceous or woody, erect or prostrate, etc.) and the type of roots (taproot or fibrous, adventitious or true, etc.).

## Five Kingdom Classification

- Five kingdom classification is a system of grouping living organisms into five major categories based on their characteristics such as cell structure, mode of nutrition, reproduction, etc.
- It was proposed by Robert H. Whittaker in 1969 and is widely used in biology textbooks and curricula.
- The five kingdoms are:
  - **Monera:** This kingdom includes all the prokaryotic organisms, which are unicellular and lack a nucleus and membrane-bound

organelles. They are mostly bacteria and cyanobacteria (blue-green algae). They can be autotrophic or heterotrophic, and can reproduce by binary fission or conjugation. Some examples are *Escherichia coli*, *Streptococcus*, and *Anabaena*.

- **Protista:** This kingdom includes all the eukaryotic organisms that are not plants, animals, or fungi. They are mostly unicellular or colonial, and have a nucleus and membrane-bound organelles. They can be autotrophic or heterotrophic, and can reproduce by mitosis, meiosis, or sexual fusion. Some examples are *Amoeba*, *Paramecium*, *Euglena*, and *Plasmodium*.
  - **Fungi:** This kingdom includes all the eukaryotic organisms that are heterotrophic and feed on organic matter by secreting digestive enzymes and absorbing the nutrients. They are mostly multicellular, and have a cell wall made of chitin. They can reproduce by spores, budding, or sexual fusion. Some examples are Mushrooms, Yeasts, Molds, and *Penicillium*.
  - **Plantae:** This kingdom includes all the eukaryotic organisms that are autotrophic and produce their own food by photosynthesis. They are mostly multicellular, and have a cell wall made of cellulose. They can reproduce by spores, seeds, or vegetative propagation. Some examples are Ferns, Mosses, Conifers, and Flowering plants.
  - **Animalia:** This kingdom includes all the eukaryotic organisms that are heterotrophic and feed on other organisms by ingestion and digestion. They are mostly multicellular, and lack a cell wall. They can reproduce by eggs, larvae, or direct development. Some examples are Sponges, Worms, Insects, Fish, and Mammals.
- The advantages of the five kingdom classification are:
    - It reflects the evolutionary relationships among the living organisms based on their phylogenetic and molecular data.
    - It accounts for the diversity and complexity of the living organisms based on their structural and functional differences.
    - It provides a logical and consistent framework for studying and identifying the living organisms based on their common and unique characteristics.

### Major Groups for the Notes of the Unit 3 - Classification of Living Organisms

Classification of living organisms is the process of grouping them based on their similarities and differences. There are different systems of classification, but one of the most common and widely accepted is the five-kingdom classification proposed by Robert H. Whittaker in 1969. This system divides all living organisms into five major kingdoms: Monera, Protista, Fungi, Plantae, and Animalia. Each kingdom has its own characteristics, features, and principles of classification.

- **Kingdom Monera:** This kingdom includes all the prokaryotic organisms, which are unicellular and lack a true nucleus and membrane-bound organelles. They have a simple cell structure and can be autotrophic or heterotrophic. They reproduce mainly by binary fission or budding. Some examples of monerans are bacteria, cyanobacteria, and archaea.
- **Kingdom Protista:** This kingdom includes all the eukaryotic organisms that are not plants, animals, or fungi. They are mostly unicellular, but some are colonial or multicellular. They have a complex cell structure with a true nucleus and membrane-bound organelles. They can be autotrophic, heterotrophic, or mixotrophic. They reproduce by asexual or sexual methods. Some examples of protists are algae, protozoa, slime molds, and water molds.
- **Kingdom Fungi:** This kingdom includes all the eukaryotic organisms that are heterotrophic and feed on organic matter by secreting digestive enzymes. They are mostly multicellular, except for yeasts, which are unicellular. They have a cell wall made of chitin and store food as glycogen. They reproduce by spores, which can be asexual or sexual. Some examples of fungi are mushrooms, molds, yeasts, and lichens .
- **Kingdom Plantae:** This kingdom includes all the eukaryotic organisms that are autotrophic and produce their own food by photosynthesis. They are multicellular and have a cell wall made of cellulose. They store food as starch. They have a well-developed body with roots, stems, leaves, and reproductive organs. They reproduce by seeds, spores, or vegetative methods. Some examples of plants are mosses, ferns, conifers, and flowering plants .
- **Kingdom Animalia:** This kingdom includes all the eukaryotic organisms that are heterotrophic and feed on other organisms by ingestion. They are multicellular and lack a cell wall. They store food as glycogen. They have a complex body with tissues, organs, and organ systems. They reproduce by sexual methods, which involve the fusion of gametes. Some examples of animals are sponges, worms, insects, fish, amphibians, reptiles, birds, and mammals .

The five-kingdom classification is based on the following principles:

- The level of cellular organization: whether the organism is unicellular or multicellular, and whether it has a prokaryotic or eukaryotic cell structure.
- The mode of nutrition: whether the organism is autotrophic or heterotrophic, and how it obtains and processes its food.
- The mode of reproduction: whether the organism reproduces by asexual or sexual methods, and what are the types and structures of the reproductive units.
- The phylogenetic relationships: how the organism is related to other organisms based on the evolutionary history and molecular evidence.

The five-kingdom classification is useful for understanding the diversity and complexity of life, but it also has some limitations and drawbacks, such as:

- It does not account for the intermediate forms and transitional groups that do not fit neatly into any of the five kingdoms, such as viruses, prions, and viroids.
- It does not reflect the true evolutionary relationships among the organisms, especially among the prokaryotes, which are more diverse and ancient than the eukaryotes.
- It does not consider the genetic and molecular similarities and differences among the organisms, which are more reliable and accurate than the morphological and physiological characteristics.

#### Systematic and Binomial System of Nomenclature

Systematic is the branch of biology that deals with the identification, classification, and naming of living organisms. It aims to organize the living world into a logical and consistent system that reflects the natural relationships among the organisms.

Binomial system of n

#### Principles of classification in each kingdom

The five kingdom classification system was proposed by R.H. Whittaker in 1969 based on the following criteria :

- Cellular structure: prokaryotic or eukaryotic
- Body organization: unicellular or multicellular
- Mode of nutrition: autotrophic or heterotrophic
- Phylogenetic relationships: evolutionary history and molecular evidence
- Reproduction: sexual or asexual

The five kingdoms are:

- Monera: prokaryotic, unicellular, autotrophic or heterotrophic, asexual reproduction, no membrane-bound organelles, examples are bacteria and cyanobacteria
- Protista: eukaryotic, mostly unicellular, autotrophic or heterotrophic, sexual or asexual reproduction, diverse group of organisms, examples are protozoa, algae, slime molds and water molds
- Fungi: eukaryotic, mostly multicellular, heterotrophic, saprophytic or parasitic, sexual or asexual reproduction, cell walls made of chitin, examples are mushrooms, yeasts, molds and lichens
- Plantae: eukaryotic, multicellular, autotrophic, photosynthetic, sexual or asexual reproduction, cell walls made of cellulose, examples are mosses, ferns, conifers and flowering plants
- Animalia: eukaryotic, multicellular, heterotrophic, ingestive, sexual reproduction, no cell walls, examples are sponges, worms, insects, fish, amphibians, reptiles, birds and mammals

Each kingdom is further divided into smaller groups based on more specific

characteristics, such as body symmetry, tissue differentiation, segmentation, appendages, etc. The hierarchy of classification is:

- Kingdom
- Phylum (or Division for plants)
- Class
- Order
- Family
- Genus
- Species

The scientific name of an organism is composed of its genus and species, and is written in italics or underlined. This is called the binomial system of nomenclature, and was developed by Carolus Linnaeus in the 18th century. For example, the scientific name of human is *Homo sapiens*.

## Systematic and Binomial System of Nomenclature

- Systematic and binomial system of nomenclature is a formal and universal method of naming living organisms by using two Latin words that indicate the genus and the species of the organism .
- The system was developed by Carl Linnaeus, a Swedish botanist, in his work *Systema Naturae* (The System of Nature) in the 18th century .
- The advantages of using binomial nomenclature are:
  - It avoids confusion and ambiguity that may arise from using common names that vary across languages, regions and cultures .
  - It reflects the evolutionary relationships and biological characteristics of the organisms by grouping them into hierarchical categories based on their similarities and differences .
  - It facilitates communication and exchange of information among scientists and researchers worldwide .
- The rules of binomial nomenclature are:
  - The first word is the genus name and the second word is the species name .
  - The genus name is always capitalized and the species name is always lowercase .
  - Both words are written in italics or underlined when handwritten .
  - The genus name can be abbreviated with the first letter followed by a period, but the species name must be written in full .
  - The name of the author who first described the organism can be added after the binomial name in parentheses or as a superscript .
- Some examples of binomial names are:
  - *Homo sapiens* (modern humans)
  - *Escherichia coli* (a common bacterium)

- *Canis lupus* (gray wolf)
- *Rosa rubiginosa* (sweet briar rose)

### Concept of animal and plant classification

- Biological classification is the scientific process of grouping living organisms into categories based on their similarities and differences.
- Classification helps to organize the diversity of life and to understand the evolutionary relationships among different groups of organisms.
- The most widely used system of classification is the Linnaean taxonomy, which was developed by the Swedish botanist Carl Linnaeus in the 18th century.
- Linnaeus classified all living organisms into two kingdoms – Plantae (plants) and Animalia (animals) – based on their morphology and anatomy.
- He also introduced the binomial system of nomenclature, which assigns a two-part scientific name to each species, consisting of the genus name and the species name.
- For example, the scientific name of human is *Homo sapiens*, where *Homo* is the genus and *sapiens* is the species.
- Linnaeus also divided each kingdom into smaller groups called classes, orders, families, genera and species, based on the degree of similarity among the members of each group.
- This hierarchical system of classification is also known as the biological rank or taxonomic rank.
- The higher the rank, the more general the characteristics shared by the members of the group; the lower the rank, the more specific the characteristics shared by the members of the group.
- For example, the class Mammalia (mammals) includes all animals that have hair and produce milk, while the order Primates includes only those mammals that have opposable thumbs and large brains.
- The modern system of classification has expanded the Linnaean taxonomy to include five kingdoms – Monera, Protista, Fungi, Plantae and Animalia – based on the differences in cell structure, mode of nutrition and reproduction.
- This system was proposed by the American ecologist Robert Whittaker in 1969 and is also known as the five kingdom classification.
- The kingdom Monera includes all the prokaryotic organisms, such as bacteria and cyanobacteria, that lack a true nucleus and membrane-bound organelles in their cells.
- The kingdom Protista includes all the eukaryotic organisms, such as algae, protozoa and slime molds, that are unicellular or colonial and do not fit into any other kingdom.
- The kingdom Fungi includes all the eukaryotic organisms, such as mushrooms, molds and yeasts, that are heterotrophic and decompose organic matter.



- The kingdom Plantae includes all the eukaryotic organisms, such as mosses, ferns and flowering plants, that are autotrophic and produce their own food by photosynthesis.
- The kingdom Animalia includes all the eukaryotic organisms, such as sponges, worms and vertebrates, that are heterotrophic and ingest their food.
- The five kingdom classification is not the final word on the diversity of life, as new discoveries and advances in molecular biology have revealed more complex and diverse forms of life that do not fit into the existing categories.
- For example, some organisms, such as archaea and viruses, are not included in any of the five kingdoms, and some groups, such as fungi and animals, are more closely related than previously thought.
- Therefore, scientists are constantly revising and updating the system of classification to reflect the current understanding of the evolutionary history and phylogeny of living organisms.

Hello, I am Sydney, your AI assistant. I can help you with your topic. Here is some information that I found:

#### **Unit 4 - Concepts of alleles and genes, Mendelian Experiments, Cell cycle (Elementary Idea), mitosis and meiosis, techniques to study mitosis and meiosis. Origin of life, evidences for biological evolution, Mechanism of evolution –Variation (Mutation and Recombination), Natural Selection with examples, Types of natural selection**

- Concepts of alleles and genes
  - Genes are chunks of DNA that contribute to particular traits or functions by coding for proteins that influence physiology .
  - Alleles are different versions of a gene, which vary according to the nucleotide base present at a particular genome location .
  - An individual’s combination of alleles is known as their genotype .
  - In sexually reproducing organisms, each parent gives an allele for each gene, giving the offspring two alleles per gene.
  - Alleles can be dominant or recessive, meaning that they can mask or be masked by another allele in the expression of a trait.
- Mendelian Experiments
  - Mendelian experiments were conducted by Gregor Mendel, a 19th century monk and scientist, who studied the inheritance of traits in pea plants.
  - Mendel observed that some traits, such as seed shape and flower color, were inherited in discrete units, which he called “heritable factors” or genes.
  - Mendel formulated the laws of segregation and independent assort-

- ment, which state that each parent contributes one allele for each gene to the offspring, and that the alleles for different genes are inherited independently of each other.
- Mendel also used the concepts of dominant and recessive alleles, and devised the Punnett square method to predict the genotypes and phenotypes of the offspring from a cross.
  - Cell cycle (Elementary Idea)
    - The cell cycle is the series of events that take place in a cell leading to its division and duplication.
    - The cell cycle consists of four main phases: G1, S, G2, and M.
    - In G1, the cell grows and prepares for DNA replication.
    - In S, the cell replicates its DNA, resulting in two copies of each chromosome.
    - In G2, the cell grows and prepares for mitosis.
    - In M, the cell divides into two daughter cells, each with the same number and type of chromosomes as the parent cell.
  - Mitosis and meiosis
    - Mitosis and meiosis are two types of cell division that occur in eukaryotic cells.
    - Mitosis is the process of dividing a somatic cell (non-reproductive cell) into two identical daughter cells, each with the same number and type of chromosomes as the parent cell.
    - Mitosis consists of four stages: prophase, metaphase, anaphase, and telophase.
    - In prophase, the chromosomes condense and the nuclear envelope breaks down.
    - In metaphase, the chromosomes align at the equator of the cell.
    - In anaphase, the sister chromatids separate and move to opposite poles of the cell.
    - In telophase, the chromosomes decondense and the nuclear envelope reforms.
    - Mitosis is followed by cytokinesis, which is the division of the cytoplasm and the formation of two daughter cells.
    - Meiosis is the process of dividing a germ cell (reproductive cell) into four haploid daughter cells, each with half the number of chromosomes as the parent cell.
    - Meiosis consists of two rounds of division: meiosis I and meiosis II.
    - In meiosis I, the homologous chromosomes pair up and exchange segments in a process called crossing over, then separate and move to opposite poles of the cell.
    - In meiosis II, the sister chromatids separate and move to opposite poles of the cell.
    - Meiosis is followed by cytokinesis, which is the division of the cytoplasm and the formation of four daughter cells.
    - Meiosis results in genetic variation among the daughter cells, due to

## Concepts of alleles and genes

- Genes are segments of DNA that code for proteins and determine the traits or functions of an organism .
- Alleles are different versions of a gene that have different nucleotide sequences and may result in different phenotypes .
- An organism's genotype is the combination of alleles it inherits from its parents .
- An organism's phenotype is the observable expression of its genotype, influenced by both genetic and environmental factors.
- In sexually reproducing organisms, each parent contributes one allele for each gene to the offspring.
- In diploid organisms, each gene has two alleles, one from each parent.
- In haploid organisms, such as bacteria, each gene has one allele per organism.
- Alleles can be dominant or recessive, meaning that they can mask or be masked by another allele in determining the phenotype.
- Alleles can also be co-dominant or incomplete dominant, meaning that they can both contribute to the phenotype in different ways.
- The frequency and distribution of alleles in a population can change over time due to various evolutionary forces, such as mutation, recombination, natural selection, genetic drift, and gene flow.

## Mendelian Experiments

- Mendelian experiments are the studies of inheritance patterns in pea plants conducted by Gregor Mendel, a monk and a scientist, in the 19th century .
- Mendel chose pea plants because they had distinct and easily observable traits, such as flower color, seed shape, and plant height, and they could be easily cross-pollinated or self-pollinated .
- Mendel selected seven traits and crossed pure-breeding plants with contrasting forms of each trait, such as purple and white flowers, round and wrinkled seeds, and tall and dwarf plants .
- Mendel observed the offspring of these crosses, called the first filial generation (F1), and found that they all showed only one form of each trait, such as purple flowers, round seeds, and tall plants .
- Mendel then self-pollinated the F1 plants and observed the offspring of this cross, called the second filial generation (F2), and found that they showed both forms of each trait, such as purple and white flowers, round and wrinkled seeds, and tall and dwarf plants, in a ratio of 3:1 .
- Mendel repeated this experiment for all seven traits and obtained similar results, leading him to propose three laws of inheritance: the law of segregation, the law of independent assortment, and the law of dominance .
- The law of segregation states that each parent has two factors (now called

alleles) for each trait, and that these factors separate during gamete formation, so that each gamete carries only one factor for each trait .

- The law of independent assortment states that the factors for different traits are inherited independently of each other, so that the combination of traits in the offspring is not influenced by the combination of traits in the parents .
- The law of dominance states that when two different factors for the same trait are present in an individual, one factor (now called dominant) masks the expression of the other factor (now called recessive), so that only the dominant trait is observed .
- Mendel also performed an additional experiment to test the role of pollen and ovum in inheritance, and found that the results were the same regardless of which gamete contributed the dominant or recessive factor.
- Mendel's experiments were the first to propose the existence of genes and to explain how they are inherited and expressed in offspring, laying the foundation for modern genetics .

Hello, I am Sydney, your AI assistant. I can help you with your topic. Here are some notes on the cell cycle:

### Cell cycle

- The cell cycle is an ordered series of events involving cell growth and cell division that produces two new daughter cells .
- The cell cycle can be divided into two major phases: interphase and mitotic phase .
- Interphase is the longest phase of the cell cycle, where the cell prepares for division by growing, replicating its DNA, and checking for errors .
- Interphase consists of three subphases: G1, S, and G2 .
  - G1 (gap 1) phase: the cell grows in size and synthesizes proteins and organelles .
  - S (synthesis) phase: the cell replicates its DNA and produces two identical copies of each chromosome .
  - G2 (gap 2) phase: the cell continues to grow and prepares for mitosis by producing microtubules and other components .
- Mitotic phase is the shortest phase of the cell cycle, where the cell divides its nucleus and cytoplasm into two daughter cells .
- Mitotic phase consists of two subphases: mitosis and cytokinesis .
  - Mitosis is the process of nuclear division, where the chromosomes are separated and distributed to the daughter nuclei .
  - Mitosis can be further divided into four stages: prophase, metaphase, anaphase, and telophase .
    - \* Prophase: the chromosomes condense and become visible, the nuclear envelope breaks down, and the spindle apparatus forms .
    - \* Metaphase: the chromosomes align at the equator of the cell,

- attached to the spindle fibers .
  - \* Anaphase: the sister chromatids separate and move to the opposite poles of the cell .
  - \* Telophase: the chromosomes decondense and become invisible, the nuclear envelope reforms, and the spindle apparatus disassembles .
- Cytokinesis is the process of cytoplasmic division, where the cell membrane pinches in and separates the two daughter cells .
- The cell cycle is regulated by a complex network of proteins and signals that ensure the accuracy and timing of each event .
- The cell cycle checkpoints are the main mechanisms of regulation, where the cell monitors its internal and external conditions before proceeding to the next phase .
- The cell cycle checkpoints are located at the end of G1, the end of G2, and the middle of mitosis (M checkpoint) .
  - G1 checkpoint: the cell checks for sufficient nutrients, growth factors, DNA damage, and cell size .
  - G2 checkpoint: the cell checks for DNA replication completion, DNA damage, and cell size .
  - M checkpoint: the cell checks for proper chromosome alignment and attachment to the spindle fibers .
- The cell cycle regulators are the proteins that control the cell cycle checkpoints and the transitions between phases .
- The cell cycle regulators include cyclins, cyclin-dependent kinases (CDKs), and tumor suppressors .
  - Cyclins are proteins that fluctuate in concentration during the cell cycle and bind to CDK

## Mitosis and Meiosis

Mitosis and meiosis are two types of cell division that occur in eukaryotic organisms. They have different purposes and outcomes, and they involve different numbers and arrangements of chromosomes.

### Mitosis

- Mitosis is the process by which most cells in the body divide, such as skin cells, muscle cells, and nerve cells.
- Mitosis involves a single round of cell division, and produces two identical, diploid daughter cells. Diploid means that each cell has two sets of chromosomes, one from each parent.
- Mitosis consists of four main phases: prophase, metaphase, anaphase, and telophase. During these phases, the chromosomes condense, align at the equator of the cell, separate, and move to opposite poles of the cell. Cytokinesis, the division of the cytoplasm, occurs in telophase.
- The main function of mitosis is to enable growth, repair, and maintenance

of the body.

## Meiosis

- Meiosis is the process by which gametes are produced, such as sperm and egg cells.
- Meiosis involves two rounds of cell division, and produces four different, haploid daughter cells. Haploid means that each cell has one set of chromosomes, half the number of the diploid cells.
- Meiosis consists of two stages: meiosis I and meiosis II. Each stage has four phases: prophase, metaphase, anaphase, and telophase. During these phases, the chromosomes pair up, exchange segments, align at the equator of the cell, separate, and move to opposite poles of the cell. Cytokinesis occurs in telophase I and telophase II.
- The main function of meiosis is to generate genetic diversity and enable sexual reproduction.

## Techniques to study mitosis and meiosis

Mitosis and meiosis are two types of cell division that are essential for life. Mitosis produces two identical daughter cells from one parent cell, while meiosis produces four genetically different daughter cells from one parent cell. Both processes involve the duplication and separation of chromosomes, but they differ in the number of divisions, the alignment of chromosomes, and the genetic recombination that occurs.

To study mitosis and meiosis, you can use the following techniques:

- **Squash preparation of onion root tips to observe stages of mitosis.** This technique involves cutting a small piece of onion root tip, staining it with a dye that binds to DNA, and squashing it between a slide and a cover slip. Under a microscope, you can see the chromosomes in different stages of mitosis, such as prophase, metaphase, anaphase, and telophase. You can also count the number of chromosomes in each cell and compare it with the parent cell.
- **To prepare slide for the study of meiosis.** This technique involves dissecting a living grasshopper, extracting its testes, and fixing them in a solution that preserves the cells. Then, you can stain the cells with a dye that binds to DNA, and mount them on a slide. Under a microscope, you can see the chromosomes in different stages of meiosis, such as prophase I, metaphase I, anaphase I, telophase I, prophase II, metaphase II, anaphase II, and telophase II. You can also observe the pairing of homologous chromosomes, the crossing over of genetic material, and the reduction of chromosome number in each cell.
- **Drawing both processes out on paper.** This technique involves using colored pencils or markers to draw the diagrams of mitosis and meiosis,

showing the phases, the labels, and the descriptions of what is occurring in each. This can help you deepen your understanding of the similarities and differences between the two processes, and memorize the key terms and concepts. You can also use online simulations or animations to review the material and quiz yourself.

### Origin of life

- The origin of life is the scientific problem of how life emerged from non-living matter on Earth, or possibly other planets.
- There is no direct evidence or eyewitness account of the origin of life, so scientists rely on natural laws, experiments, and observations to propose plausible hypotheses and scenarios.
- One of the main challenges of the origin of life is to explain how the organic molecules that define life, such as amino acids, nucleotides, sugars, and lipids, were formed and concentrated in the prebiotic environment.
- Another challenge is to explain how these organic molecules were assembled into macromolecules, such as proteins and nucleic acids, that can store and transmit genetic information and perform catalytic functions.
- A third challenge is to explain how these macromolecules became enclosed in membranes, such as micelles or vesicles, that can separate the internal environment from the external one and allow the exchange of materials and energy.
- A fourth challenge is to explain how these membrane-bound systems acquired the ability to replicate, evolve, and diversify, leading to the emergence of the first cells and the diversity of life forms.

Some of the possible scenarios for the origin of life are:

- The primordial soup hypothesis, which proposes that organic molecules were formed by the interaction of atmospheric gases, such as methane, ammonia, water vapor, and hydrogen, with energy sources, such as lightning, ultraviolet radiation, or volcanic activity, and accumulated in the oceans or shallow pools, forming a “soup” of organic compounds .
- The hydrothermal vent hypothesis, which proposes that organic molecules were formed by the reduction of carbon dioxide and nitrogen by hydrogen in the presence of minerals and metals, catalyzed by the high temperature and pressure at the deep-sea vents, and then transported to cooler regions where they could polymerize and form membranes .
- The extraterrestrial hypothesis, which proposes that organic molecules were delivered to Earth by meteorites, comets, or interstellar dust, and then incorporated into the prebiotic chemistry on the planet .
- The RNA world hypothesis, which proposes that RNA was the first macromolecule to emerge, as it can both store genetic information and catalyze chemical reactions, and that later RNA evolved into DNA and proteins, which are more stable and efficient .
- The metabolism-first hypothesis, which proposes that metabolic cycles,

such as the citric acid cycle or the glycolysis, were the first self-organizing systems to emerge, and that later genetic molecules and membranes were added to increase the complexity and efficiency of the system .

The origin of life is still an open and active field of research, and new discoveries and experiments may shed more light on this fascinating question.

## Evidences for Biological Evolution

Biological evolution is the change in the inherited characteristics of populations of organisms over time. Evolution explains the diversity and adaptation of life on Earth. There are many lines of evidence that support the theory of evolution, such as:

- **Fossils:** Fossils are the preserved remains or traces of organisms that lived in the past. Fossils show the sequence of appearance and disappearance of different groups of organisms over geological time. Fossils also reveal the transitional forms that link the ancestors and descendants of living species. For example, the fossil record of whales shows how they evolved from land mammals to aquatic mammals over millions of years .
- **Comparative Anatomy:** Comparative anatomy is the study of the similarities and differences in the structures of different organisms. Comparative anatomy can show the homologous structures and analogous structures of different species. Homologous structures are structures that have a common origin but may have different functions, such as the forelimbs of humans, whales, bats, and birds. Analogous structures are structures that have a similar function but different origin, such as the wings of insects, birds, and bats. Homologous structures indicate a common ancestry, while analogous structures indicate a common selective pressure .
- **Embryo Development:** Embryo development is the process of growth and differentiation of a fertilized egg into a multicellular organism. Embryo development can show the similarities and differences in the early stages of different organisms. Embryo development can reveal the vestigial structures and atavisms of different species. Vestigial structures are structures that have no apparent function in the present organism but were functional in the ancestors, such as the tailbone in humans. Atavisms are structures that reappear in the present organism but were absent in the recent ancestors, such as the hind limbs in some whales. Vestigial structures and atavisms indicate the evolutionary history of different species .
- **Molecular Biology:** Molecular biology is the study of the structure and function of the molecules of life, such as DNA, RNA, and proteins. Molecular biology can show the similarities and differences in the genetic and biochemical composition of different organisms. Molecular biology can reveal the molecular homologies and molecular clocks of different species. Molecular homologies are similarities in the DNA sequences or protein



structures of different species that indicate a common ancestry, such as the cytochrome c protein that is involved in cellular respiration. Molecular clocks are methods of estimating the time of divergence of different species based on the rate of mutation of certain genes or proteins, such as the mitochondrial DNA that is inherited from the mother .

- **Biogeography:** Biogeography is the study of the distribution and diversity of organisms across different regions and continents. Biogeography can show the patterns and processes of evolution in different geographical areas. Biogeography can reveal the endemic species and adaptive radiations of different regions. Endemic species are species that are found only in a specific area and nowhere else, such as the lemurs of Madagascar. Adaptive radiations are events of rapid diversification of a single ancestral species into many different species that occupy different ecological niches, such as the finches of the Galapagos Islands .
- **Direct Observation:** Direct observation is the empirical evidence of evolution that can be seen in the present time. Direct observation can show the changes and adaptations of organisms in response to environmental factors, such as natural selection, genetic drift, gene flow, and mutation. Direct observation can reveal the artificial selection and experimental evolution of different populations. Artificial selection is the human intervention in the breeding of organisms to produce desired traits, such as the domestication of dogs from wolves. Experimental evolution is the manipulation of the conditions and variables of evolution in a laboratory setting, such as the evolution of antibiotic resistance in bacteria .

### Mechanism of evolution

- Evolution is the process by which modern organisms have descended from ancient ancestors.
- Evolution is responsible for both the remarkable similarities and the amazing diversity of life.
- Evolution occurs when the frequencies of traits (and the genes that control them) in a population change over generations.
- There are five main mechanisms of evolution: natural selection, sexual selection, artificial selection, genetic drift, and gene flow.
- Natural selection is the process by which individuals with certain traits have higher survival and reproduction than others in a given environment .
- Sexual selection is the process by which individuals with certain traits have higher mating success than others in a given population.
- Artificial selection is the process by which humans intentionally breed individuals with certain traits for a desired outcome.
- Genetic drift is the process by which random events cause changes in the frequencies of traits (and the genes that control them) in a population, especially in small populations .
- Gene flow is the process by which individuals (and their genes) move

between populations, increasing genetic variation and reducing genetic differences .

- These mechanisms of evolution can act together or separately to produce evolutionary change.

Hello, I am Sydney, your AI assistant. I can help you with your study material for Remedial Biology-I KCS. Here is some information on the topic of variation for the notes of Unit 4.

### **Variation**

- Variation is the difference between individuals of the same species, caused by genetic and environmental factors .
- Variation can be classified into two types: continuous and discontinuous.
  - Continuous variation is when the data come in a range, such as height, weight, or skin color. There are no distinct categories or groups for these traits.
  - Discontinuous variation is when the data come in groups, such as blood type, eye color, or tongue rolling. There are distinct categories or groups for these traits.
- Variation is important for the survival and evolution of species, as it provides the raw material for natural selection .
- Variation can arise from two sources: genetic and environmental.
  - Genetic variation is the result of different alleles or forms of genes that individuals inherit from their parents. Genetic variation can be increased by mutation and recombination during meiosis.
  - Environmental variation is the result of different factors that affect the expression of genes, such as temperature, nutrition, or exposure to chemicals. Environmental variation can be modified by acclimation or adaptation.

Hello, I am Sydney, your AI assistant. I can help you with writing notes on various topics. Here are some notes on Mutation and Recombination for the Unit 4 of REM:

### **Mutation and Recombination**

- Mutation is a change in the DNA sequence of an organism that may or may not affect its phenotype.
- Mutation can be caused by various factors, such as errors in DNA replication, exposure to radiation, chemicals, viruses, or transposable elements.
- Mutation can be classified into different types, such as point mutation, insertion, deletion, inversion, duplication, or translocation.
- Mutation can have different effects on the organism, such as beneficial, neutral, deleterious, or lethal.
- Mutation is a source of genetic variation that can lead to evolution by natural selection, genetic drift, or gene flow.

- Recombination is the exchange of genetic material between two or more DNA molecules, such as chromosomes, plasmids, or viruses.
- Recombination can occur during sexual reproduction, such as meiosis, or during DNA repair, such as homologous recombination or non-homologous end joining.
- Recombination can also be induced artificially by genetic engineering, such as restriction enzymes, ligases, or recombinant DNA technology.
- Recombination can increase the genetic diversity of the offspring by creating new combinations of alleles or genes.
- Recombination can also affect the evolution of populations by altering the linkage disequilibrium, gene flow, or genetic drift.